ApproxDA-TransUNet: A Lightweight Dual-Attention Architecture for Medical Image Segmentation

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Abstract

Dual-attention mechanisms jointly model long-range spatial (position) and channel dependencies and perform strongly in medical image segmentation, but their quadratic complexity and the heavy hybrid CNN-Transformer backbones they are built on make networks such as DA-TransUNet costly (≈108M parameters and ≈30 GFLOPs per 224×224 slice). We present **ApproxDA-TransUNet**, a lightweight dual-attention architecture that approximates both attention branches and slims the network around them. Each ApproxDA block integrates windowed low-rank position attention (window size $M = 28$, rank $r = 32$) with grouped channel attention ($G = 8$); the blocks sit on the bottleneck and on all three skip connections of a network built from a ResNet-50 stem truncated at stride 8, a LightViT-T bottleneck transformer, and a slim decoder. The model uses 90.0% fewer parameters (10.75M vs. 107.95M) and 77.1% fewer FLOPs (6.92 vs. 30.21 GFLOPs) than DA-TransUNet. In single-axis ablations on Synapse, removing the ApproxDA blocks reduces Dice by 2.76 points ($p = 0.008$), and the approximation costs no accuracy: exact global dual attention, as in DA-TransUNet, gives the same Dice (−0.01 points) at 14% more FLOPs and twice the training time. Across five benchmarks (Synapse CT, ACDC MRI, CVC-ClinicDB colonoscopy, ISIC 2018 dermoscopy, and FeTA 2.4 fetal MRI) with three seeds each, ApproxDA-TransUNet retains the Dice of DA-TransUNet: its three-seed mean is higher on all five datasets (average gain +0.37 points, 10 of 15 seed pairs positive), although no single dataset shows a significant

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difference and HD95 is higher on both datasets for which it is reported.

Keywords: Medical image segmentation, Dual attention, Attention approximation, Lightweight architecture, Hybrid CNN-Transformer

1. Introduction

Automated biomedical image segmentation is fundamental to computer-assisted diagnosis, radiotherapy planning, and surgical navigation [1, 2, 3]. Clinical modalities span a broad spectrum—from 3D volumetric CT and MRI to 2D endoscopic video and high-resolution dermoscopy. Accurate boundary delineation across these domains requires architectures capable of both fine-grained edge localization and expansive global context modeling.

While convolutional neural networks (CNNs) [1, 2, 4] form the conventional baseline, their localized receptive fields limit long-range context aggregation. Hybrid CNN-Transformer frameworks like TransUNet [3] address this via self-attention, and dual-attention networks [5] concurrently capture spatial affinities via Position Attention Modules (PAM) and cross-channel correlations via Channel Attention Modules (CAM). Recently, DA-TransUNet [6] integrated dual-attention blocks across multi-scale skip connections and the bottleneck, demonstrating state-of-the-art accuracy on multi-organ CT and colon polyp benchmarks.

Despite their representational strength, standard dual-attention architectures suffer from severe computational overhead. Full-map PAM scales quadratically with spatial tokens ($\mathcal{O}(N^2C)$), while CAM scales quadratically with channel depth ($\mathcal{O}(C^2N)$). Coupled with a ViT-B/16 encoder and a wide decoder, DA-TransUNet requires **107.95M parameters** and **30.2 GFLOPs** per 224×224 slice (39.8 GFLOPs at 256×256), which limits its use on resource-constrained hardware. The quadratic cost of attention has prompted efficient approximations—windowed attention [7], low-rank factorization [8], and grouped channel interaction [9]—but in DA-TransUNet the dual-attention blocks account for only 2.68M parameters and 0.87 GFLOPs; most of the cost lies in the ViT-B/16 encoder and the wide decoder around them.

In our preliminary workshop study [10], we introduced the ApproxDA block, which keeps the DANet head of DA-TransUNet and replaces its attention operators with three independently controllable approximations: windowed position attention (window size M), low-rank projection (rank r),

and grouped channel attention (groups G). In that study, exact attention showed no consistent advantage over its low-rank approximation, but the model kept the R50-ViT-B/16 backbone, so the savings of the approximated attention barely reached the whole network. In this work, we carry the approximated dual attention into a lightweight architecture. We present **ApproxDA-TransUNet**, a system-level redesign of DA-TransUNet built from four components: (i) a *ResNet-50 stem truncated after its second stage* (stride 8, 512 channels) followed by a compact Vision Transformer (*LightViT-T* [11]), which reduces the encoder from 97.9M to 9.3M parameters; (ii) *approximated dual-attention blocks*, in which position attention is restricted to local windows ($M = 28$) and projected to a low rank ($r = 32$), channel attention is computed in groups ($G = 8$), and the two branches are blended with a fixed gate ($g = \frac{1}{2}$); (iii) these blocks on all three skip connections (1/8, 1/4, 1/2) and on the bottleneck after the transformer, whereas DA-TransUNet applies dual attention to the 1/8 and 1/4 skips and to the patch embeddings before the ViT; and (iv) a *slim decoder* with additive skip fusion and 64 channels at every skip stage. Together, these changes reduce the parameters from **107.95M to 10.75M** (-90.0%) and the compute from **30.21 to 6.92 GFLOPs** (-77.1%).

In single-axis ablations on Synapse, removing the dual-attention blocks reduces Dice by 2.76 points, and their approximation costs no accuracy: replacing the approximated attention by the exact global dual attention of DA-TransUNet leaves Dice unchanged while increasing FLOPs by 14% and doubling the training time.

We evaluate ApproxDA-TransUNet across five multi-modal benchmarks under a three-seed protocol (1234, 0, 1): Synapse CT, ACDC cardiac MRI, CVC-ClinicDB colonoscopy, ISIC 2018 dermoscopy, and FeTA 2.4 fetal brain MRI. ApproxDA-TransUNet retains the Dice of DA-TransUNet: its three-seed mean Dice is higher on all five datasets (average gain +0.37 percentage points, 10 of 15 seed pairs positive), although no single dataset shows a significant difference, and HD95 is higher than DA-TransUNet's on both datasets for which it is reported. On FeTA 2.4, the 10.75M model is 1.8–2.5 points above the reproduced U-Net family and Swin-Unet and within 0.1 points of TransUNet.

The primary contributions of this paper are:

- **Lightweight dual-attention architecture**: ApproxDA-TransUNet uses 90% fewer parameters and 77% fewer FLOPs than DA-TransUNet

at comparable segmentation accuracy.

- **Dual-attention approximation**: windowed low-rank position attention (M, r) and grouped channel attention (G) as three independently parameterized axes, with single-axis ablations on Synapse showing that removing the blocks reduces Dice by 2.76 points and that the approximation matches exact dual attention at lower cost.
- **Five-dataset, three-seed validation**: paired comparisons with DA-TransUNet on benchmarks spanning CT, MRI, colonoscopy, dermoscopy, and fetal MRI, with literature and reproduced baselines and a deployment profile.

The rest of this paper is organized as follows: Section 2 reviews related work; Section 3 presents the methodology; Section 4 reports experimental results; Section 5 provides ablation studies; Section 6 discusses augmentation, deployment cost, and limitations; Section 7 concludes the paper.

2. Related Work

2.1. Architectures for Biomedical Image Segmentation

U-Net [1] established the dominant encoder-decoder topology for medical image segmentation, using multi-scale skip connections to preserve high-resolution spatial details. Subsequent variants enhanced representation capacity through nested dense skips (UNet++ [2]), additive attention gates (Attention U-Net [4]), and multi-resolution residual blocks (MultiResUNet [12]). However, purely convolutional models remain constrained by localized receptive fields, hindering the disambiguation of contiguous anatomical structures.

To overcome this, Vision Transformers (ViTs) [13] were adopted for global context aggregation. TransUNet [3] pioneered hybrid architectures by coupling a ResNet-50 CNN with a ViT-B/16 bottleneck transformer. Swin-Unet [14] explored a purely transformer-based U-shaped structure with shifted local windows, while UNETR [15] and SwinUNETR [16] extended transformers to volumetric 3D segmentation. While effective, hybrid models such as TransUNet exceed 100M parameters, imposing substantial memory and computational overhead.

2.2. Dual-Attention Mechanisms and Computational Bottlenecks

Dual attention was formalized by Fu et al. [5] (DANet) to simultaneously model spatial dependencies via Position Attention Modules (PAM) and cross-channel interdependencies via Channel Attention Modules (CAM). In biomedical imaging, DA-TransUNet [6] embedded dual-attention blocks across the bottleneck and skip connections of TransUNet, establishing state-of-the-art results on multi-organ CT and polyp segmentation. Related efforts explored dual attention in hierarchical transformers (DAE-Former [17], TransDAE [18]) and state-space architectures (DAMamba [19]). Contextual attention networks such as SECA-Net [20] add squeezed-and-excitated multi-scale atrous convolution and spatial-channel attention modules to the bottleneck of a ResNet-based U-Net to strengthen context modeling. Our work takes a complementary, efficiency-oriented view: we keep the dual-attention formulation and show that its spatial and channel attention can be approximated aggressively without a measurable loss in accuracy (Section 5).

However, standard dual attention exhibits quadratic complexity: spatial self-attention scales as $\mathcal{O}(N^2C)$ for $N = H \times W$ tokens, while channel attention scales as $\mathcal{O}(C^2N)$. In multi-scale architectures, generating full affinity matrices across multiple high-resolution skip stages (112×112 and 56×56) makes memory grow quadratically with the number of tokens.

2.3. Efficient Attention Approximation and Positioning

To mitigate quadratic complexity, diverse approximation paradigms have emerged in general vision: local window partitioning (Swin Transformer [7]), low-rank key-value projection (Linformer [8], Performer [21]), channel group factorizations (EMCAD [9]), and efficient transformer backbones (LightViT [11]).

Efficient medical image segmentation has also been approached from other, complementary directions. On the decoder side, Chen et al. [22] select decoder blocks adaptively and reweight filters to bypass redundant blocks and obtain a lightweight model for each dataset. Other recent architectures capture multi-scale or global context at lower cost without full transformer attention; FE-MAANet [23], for example, combines adaptive large-kernel modules built from cascaded depthwise-separable convolutions with a frequency-spatial parallel attention module in the skip connections, and EMCAD [9] designs an efficient multi-scale attention decoder. In contrast, our work starts from a dual-attention Transformer U-Net (DA-TransUNet) and jointly simplifies the backbone, the spatial and channel attention operators, the skip processing, and the decoder within one fixed architecture, without

architecture search or auxiliary routing. Our earlier workshop study [10] introduced the approximated dual-attention block within the heavy R50-ViT-B/16 model; this work carries it into a lightweight architecture and shows that a 90% parameter reduction is possible at comparable accuracy.

3. Methodology

This section describes ApproxDA-TransUNet: the overall topology, the attention approximation in the ApproxDA block, the slim decoder, and the computational cost.

3.1. Overall Lightweight Architecture

Figure 1a shows the complete architecture. ApproxDA blocks sit on all three skip connections (1/8, 1/4, 1/2) and on the bottleneck; the value C' given for each block in the figure is its compressed width, defined below. Compared with the 108M-parameter DA-TransUNet [6], every stage is redesigned for efficiency while keeping the multi-scale skip structure. DA-TransUNet applies dual attention to the 1/8 and 1/4 skips and to the patch embeddings before the ViT; as in our workshop model [10], all three skips are equipped here, and the bottleneck block follows the transformer instead of preceding it (the two placements are compared in Section 5).

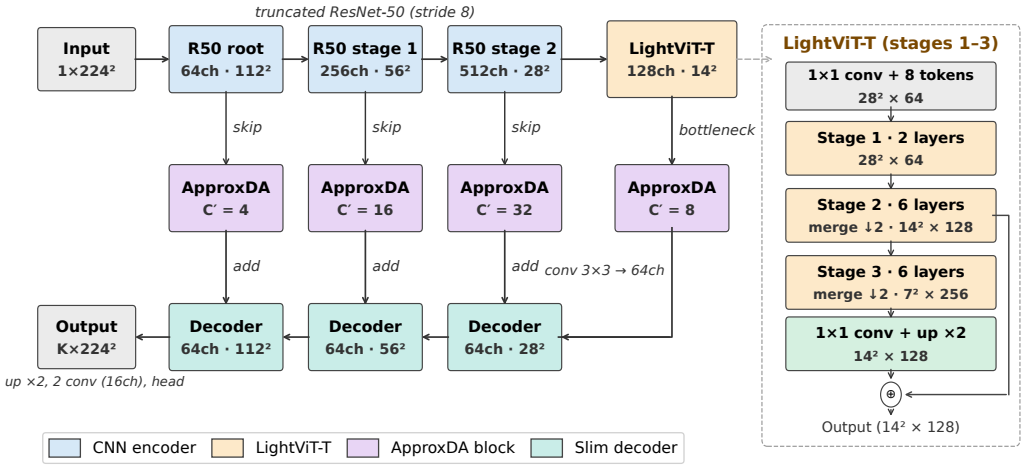
3.1.1. Early-Truncated CNN Stem

In TransUNet and DA-TransUNet, the hybrid encoder runs the first three stages of ResNet-50 (down to stride 16 and 1024 channels; 11.9M parameters, 4.0 GFLOPs). The resulting 14×14 feature map is projected to 768-dimensional tokens and processed by a 12-layer ViT-B/16 (85.1M parameters, 17.4 GFLOPs).

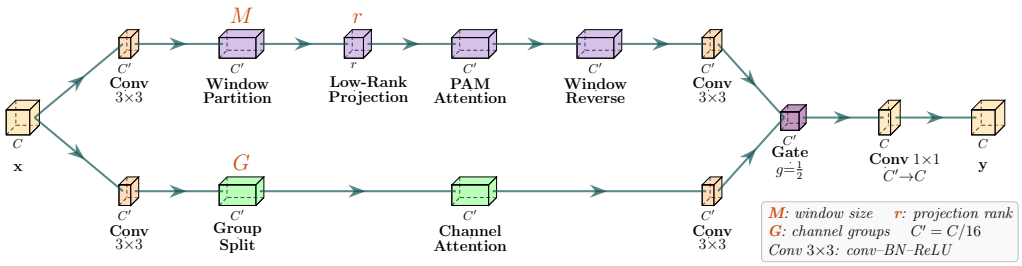
ApproxDA-TransUNet truncates ResNet-50 after its second stage. For an input image $\mathbf{I} \in \mathbb{R}^{3\times H\times W}$ (typically 224×224 or 256×256):

- **Stem & Stage 1**: Generates feature maps $\mathbf{F}_{1/2} \in \mathbb{R}^{64\times \frac{H}{2}\times \frac{W}{2}}$ (stride 2) and $\mathbf{F}_{1/4} \in \mathbb{R}^{256\times \frac{H}{4}\times \frac{W}{4}}$ (stride 4).
- **Stage 2**: Produces feature map $\mathbf{F}_{1/8} \in \mathbb{R}^{512\times \frac{H}{8}\times \frac{W}{8}}$ (stride 8).

The third stage is removed, so the CNN stem has 1.45M parameters (1.83 GFLOPs).



(a) Overall architecture.



(b) ApproxDA block.

Figure 1: ApproxDA-TransUNet.

3.1.2. LightViT-T Bottleneck and Attention Placement

Instead of ViT-B, $\mathbf{F}_{1/8}$ is projected by a 1×1 convolution and processed by the three stages of LightViT-T [11] (ImageNet-1k weights), a hierarchical lightweight vision transformer. The second-stage output and the upsampled third-stage output are fused into the bottleneck representation $\mathbf{Z} \in \mathbb{R}^{128 \times \frac{H}{16} \times \frac{W}{16}}$ (14×14 for 224×224 input). LightViT-T has 7.81M parameters but only 0.75 GFLOPs, because it operates on 28×28 and smaller token grids. The bottleneck ApproxDA block is positioned post-transformer, directly operating on $\mathbf{Z}$ to refine global representations before decoding.

3.2. The ApproxDA Block Formulation

Figure 1b illustrates the internal topology of the ApproxDA block. As in our workshop study [10], the block keeps the DANet head layout of DA-TransUNet and replaces only its attention operators. The module receives input tensor $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$ and splits computation into two parallel streams, Low-Rank Windowed Position Attention (PAM) and Grouped Channel Attention (CAM); each stream compresses the channels, applies the approximated attention, and refines the result with a 3×3 convolution, BN, and ReLU, after which the two streams are blended and projected back to C channels.

The approximation is parameterized along three independent axes: the **spatial scope** of position attention (window size M), its **representation fidelity** (projection rank r), and the span of **channel interaction** (number of groups G). Each axis can be changed while the others are held fixed, which the single-axis ablations in Section 5 use; with $r = 0$, $M \geq H, W$, and $G = 1$, the block reduces to the DANet head of DA-TransUNet, up to the factor $\frac{1}{2}$ of the fixed blend, which the following 1×1 convolution absorbs.

3.2.1. Bottleneck Convolutional Compression

As in the DANet head used by DA-TransUNet, each branch first compresses $\mathbf{X}$ with its own convolution:

$$\mathbf{X}' = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{3\times3}(\mathbf{X}))) \in \mathbb{R}^{C' \times H \times W}, \tag{1}$$

where $C' = C/16$. For the bottleneck and the 1/8, 1/4, and 1/2 skips, C' is 8, 32, 16, and 4.

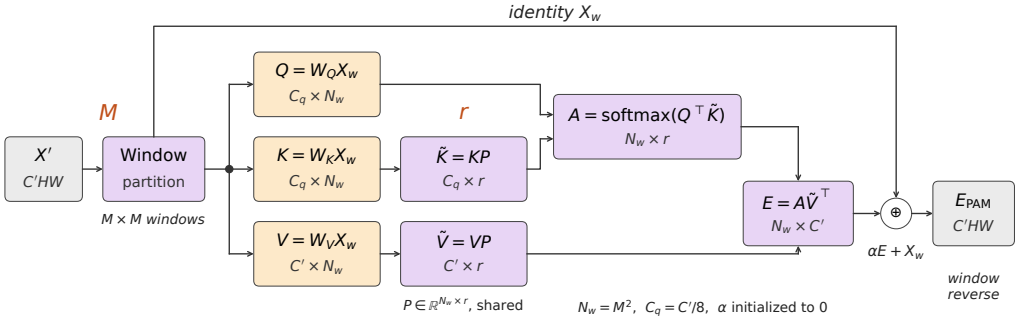


Figure 2: Low-rank windowed position attention (one window).

3.2.2. Low-Rank Windowed Position Attention (PAM)

Figure 2 shows the computation inside one window. Full spatial self-attention requires pairwise affinities between all $N = HW$ tokens, yielding $\mathcal{O}(N^2C')$ complexity. We approximate this along two orthogonal dimensions: **spatial windowing** (M) and **low-rank projection** (r).

Window Partitioning. The feature map $\mathbf{X}'$ is partitioned into non-overlapping spatial windows of size $M \times M$, where $M = \min(M, H, W)$ ($M = 28$ for 224×224 inputs). Consequently, the $1/2$ and $1/4$ skip stages are partitioned into 4×4 and 2×2 window grids, while the $1/8$ skip and bottleneck operate within a single global window. Within each window of $N_w = M^2$ tokens, queries $\mathbf{Q}$, keys $\mathbf{K}$, and values $\mathbf{V}$ are generated via 1×1 convolutions:

$$\mathbf{Q} = \mathbf{W}_Q \mathbf{X}_w, \quad \mathbf{K} = \mathbf{W}_K \mathbf{X}_w \in \mathbb{R}^{C_q \times N_w}, \quad \mathbf{V} = \mathbf{W}_V \mathbf{X}_w \in \mathbb{R}^{C' \times N_w}, \tag{2}$$

where $C_q = \max(1, \lfloor C'/8 \rfloor)$.

Low-Rank Subspace Projection. Rather than computing an $N_w \times N_w$ attention matrix, keys and values are projected into a subspace of rank r ($r = 32$) via shared projection matrix $\mathbf{P} \in \mathbb{R}^{N_w \times r}$:

$$\tilde{\mathbf{K}} = \mathbf{K}\mathbf{P} \in \mathbb{R}^{C_q \times r}, \quad \tilde{\mathbf{V}} = \mathbf{V}\mathbf{P} \in \mathbb{R}^{C' \times r}. \tag{3}$$

The windowed position attention matrix $\mathbf{A} \in \mathbb{R}^{N_w \times r}$ and refined features $\mathbf{E}_{\text{PAM}}$ are then computed as:

$$\mathbf{A} = \text{softmax}\left(\mathbf{Q}^\top \tilde{\mathbf{K}}\right), \tag{4}$$

$$\mathbf{E}_{\text{PAM}} = \alpha \left(\mathbf{A}\tilde{\mathbf{V}}^\top\right)^\top + \mathbf{X}_w \in \mathbb{R}^{C' \times N_w}, \tag{5}$$

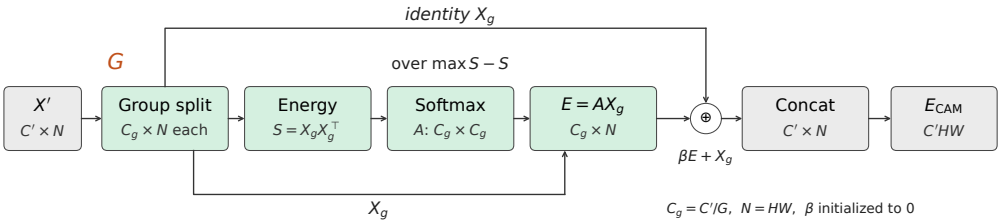


Figure 3: Grouped channel attention (one group).

where α is a learnable scalar initialized to 0. All window features are reassembled into the $H \times W$ grid and refined by a 3×3 convolution, BN, and ReLU. Setting $r = 0$ bypasses projection, recovering exact attention within each window.

3.2.3. Grouped Channel Attention (CAM)

Figure 3 shows the computation for one group. Standard CAM computes channel covariance across all C' channels with $\mathcal{O}(C'^2 N)$ complexity. ApproxDA blocks parameterize channel interactions via grouping parameter G: the C' channels are partitioned into G disjoint groups of width $C_g = C'/G$:

$$\mathbf{X}' = \left[\mathbf{X}^{(1)}, \mathbf{X}^{(2)}, \ldots, \mathbf{X}^{(G)}\right], \quad \mathbf{X}^{(g)} \in \mathbb{R}^{C_g \times N}. \tag{6}$$

Within each group g, the channel correlation energy between channel slice i and j is $s_{ij} = \mathbf{X}_i^{(g)} \cdot \mathbf{X}_j^{(g)}$. The channel attention map and modulated features are given by:

$$x_{ij}^{(g)} = \text{softmax}_j \left(\max_k s_{ik} - s_{ij} \right), \tag{7}$$

$$\mathbf{E}_{\text{CAM},i}^{(g)} = \beta \sum_{j=1}^{C_g} x_{ij}^{(g)} \mathbf{X}_j^{(g)} + \mathbf{X}_i^{(g)}, \tag{8}$$

where β is a learnable scalar initialized to 0. Groups are concatenated along the channel dimension to form $\mathbf{E}_{\text{CAM}} \in \mathbb{R}^{C' \times H \times W}$ and refined by a 3×3 convolution, BN, and ReLU; $G = 1$ recovers the original CAM. When G does not divide C', the largest divisor of C' not exceeding G is used. Because C' is small in the lightweight model, $G = 8$ leaves a single channel per group in the bottleneck ($C' = 8$) and in the 1/2 skip ($C' = 4$, four groups), where CAM reduces to $(1 + \beta)\mathbf{X}'$; the cost of CAM is negligible at every stage. $G = 8$ is therefore a fixed setting retained from the workshop design rather than a tuned choice; un-grouped CAM ($G = 1$) is evaluated in Section 5.

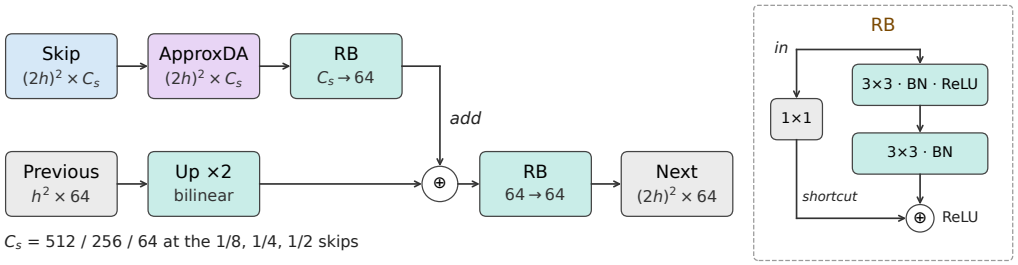


Figure 4: Slim decoder stage.

3.2.4. Fixed Equal-Blend Routing

In our earlier study [10], a learnable per-channel gate $\mathbf{g} = \sigma(\mathbf{W}_g\,\mathrm{GAP}(\mathbf{x}))$ stayed at a near-uniform blend throughout training on Synapse (block means within 0.499–0.512 over 300 epochs) and did not outperform single-branch routing. ApproxDA-TransUNet therefore fixes the gate to an equal blend:

$$\mathbf{Y} = \mathbf{W}_f\,(0.5 \cdot \mathbf{E}_{\mathrm{PAM}} + 0.5 \cdot \mathbf{E}_{\mathrm{CAM}}) \in \mathbb{R}^{C\times H\times W}, \tag{9}$$

where $\mathbf{W}_f$ consists of channel dropout ($p = 0.05$), a 1×1 convolution back to C channels, and ReLU, as in the DANet head; there is no outer residual connection.

3.3. Compact Decoder Architecture

Unlike the concatenation decoder of DA-TransUNet ($256 \to 128 \to 64 \to 16$ channels), ApproxDA-TransUNet uses 64 channels at every skip stage and fuses skips by addition (Figure 4). A 3×3 convolution first maps $\mathbf{Z}$ to 64 channels. At each skip stage $l \in \{1, 2, 3\}$ (1/8, 1/4, 1/2), the previous map is bilinearly upsampled by $2\times$ and added to the ApproxDA-refined skip $\mathbf{S}_l$ after a channel-reduction block:

$$\mathbf{D}_l = \mathrm{RB}\left(\mathrm{Upsample}_{2\times}(\mathbf{D}_{l-1}) + \mathrm{RB}(\mathbf{S}_l)\right) \in \mathbb{R}^{64\times H_l\times W_l}, \tag{10}$$

where RB is a residual block of two 3×3 convolutions with batch normalization and a 1×1 shortcut. The full-resolution stage upsamples once more and applies two 3×3 convolutions with 16 channels, followed by a 1×1 segmentation head. The decoder without the skip blocks has 0.95M parameters (3.68 GFLOPs).

Table 1: Computational cost.

Component	DA-TransUNet [6]		ApproxDA-TransUNet (Ours)	
	Params (M)	**FLOPs (G)**	**Params (M)**	**FLOPs (G)**
Parameters and FLOPs by component				
CNN stem	11.90 (R50, 3 stages)	3.96	1.45 (R50, 2 stages)	1.83
Token embedding	0.94	0.15	–	–
Bottleneck transformer	85.06 (ViT-B/16)	17.37	7.81 (LightViT-T)	0.75
Dual-attention blocks	2.68 (DANet heads)	0.87	0.54 (4 ApproxDA blocks)	0.60
Decoder (without attention blocks)	7.39	7.78	0.95	3.68
Segmentation head	< 0.01	0.07	< 0.01	0.07
Total	**107.95**	**30.21**	**10.75**	**6.92**
Relative	100%	100%	**10.0%**	**22.9%**
Measured on the GPU				
Peak training memory (batch 24)	11.52 GiB		**4.85 GiB** (−58%)	
Peak inference memory (batch 1)	2.59 GiB		**1.09 GiB** (−58%)	
Throughput, batch 32 (slices/s)	555		**1415** (2.55×)	
Latency, batch 1	**6.5 ms**		9.6 ms	

224×224 input, 9 classes. Upper part: fvcore on CPU, attention matrix products included; DA-TransUNet instantiates three DANet heads per decoder stage but executes one, and its parameter count includes all of them. Lower part: NVIDIA RTX 5090, fp32, idle GPU; memory is the peak allocated by PyTorch, including weights and cuDNN workspace, with training at batch size 24 and SGD.

3.4. Computational Complexity and Parameter Breakdown

As detailed in the upper part of Table 1, ApproxDA-TransUNet reduces the parameters from 107.95M to **10.75M** (−90.0%) and the FLOPs from 30.21G to **6.92G** (−77.1%). Most of the saving comes from replacing ViT-B/16 and from the slim decoder; the attention blocks themselves are a small part of either model. For high-resolution skip features (112×112, $N = 12{,}544$), windowing ($M = 28$) and low-rank projection ($r = 32$) cut spatial affinity entries from $N^2 = 1.57 \times 10^8$ to $N \cdot r = 4.01 \times 10^5$ (a 392× reduction). Channel grouping divides the size of the CAM affinity matrix by G, but CAM is already negligible after the $C/16$ compression.

3.5. Optimization Objective

All models are trained with a composite objective combining Cross-Entropy ($\mathcal{L}_{\text{CE}}$) and multi-class soft Dice loss ($\mathcal{L}_{\text{Dice}}$) with equal weighting:

$$\mathcal{L} = 0.5 \cdot \mathcal{L}_{\text{CE}} + 0.5 \cdot \mathcal{L}_{\text{Dice}}. \tag{11}$$

No auxiliary losses or extra heads are used.

4. Experiments and Results

This section details the experimental protocol on five clinical benchmarks, compares ApproxDA-TransUNet on each benchmark with published or reproduced baselines and with DA-TransUNet retrained under the same protocol, summarizes the paired comparison across datasets, and shows qualitative results.

4.1. Experimental Setup and Benchmark Datasets

Models are evaluated across **five clinical benchmarks** spanning four distinct imaging modalities under a strict **three-seed protocol** (seeds: 1234, 0, 1):

1. **Synapse Multi-Organ CT** [24]: 30 volumetric abdominal CT scans (18 train, 12 test; 2211 slices) covering 8 abdominal organs (aorta, gallbladder, left/right kidneys, liver, pancreas, spleen, stomach), trained for 300 epochs at 224×224 and evaluated with Dice (%) and HD95 in voxels (the TransUNet evaluation code, without voxel spacing).
2. **ACDC Cardiac Cine-MRI** [25]: 100 patient exams (40 end-diastolic/systolic 3D test volumes) targeting right ventricle, myocardium, and left ventricle, trained for 300 epochs at 224×224 and evaluated with Dice (%).
3. **CVC-ClinicDB Colonoscopy** [26]: 612 colonoscopy frames (75/25 split) for binary polyp segmentation, trained with Adam (10^{-3}) for 500 epochs at 256×256, evaluated with Dice (%) and IoU (%).
4. **ISIC 2018 Dermoscopy** [27]: 2594 dermoscopic images (75/25 split) for binary skin-lesion segmentation, trained with SGD for 50 epochs at 256×256, evaluated with Dice (%) and IoU (%).
5. **FeTA 2.4 Fetal Brain MRI** [28]: 80 reconstructed T2-weighted fetal brain MRI volumes under the standard EffiDec3D split (64 train, 8 val, 8 test) across 7 brain tissues (eCSF, GM, WM, LV, CBM, SGM, BS), trained for 61 epochs at 224×224 and evaluated with Dice (%) and HD95 (mm, using each subject's voxel spacing).

Implementation Protocols. All models are trained with the TransUNet augmentations (random 90° rotations with flips, and small rotations), a $0.5\,\mathcal{L}_{\text{CE}} + 0.5\,\mathcal{L}_{\text{Dice}}$ loss, batch size 24, and SGD (learning rate 0.01, momentum 0.9, weight decay 10^{-4}, polynomial decay), except CVC-ClinicDB (Adam, 10^{-3}); on FeTA 2.4, Swin-Unet uses its official learning rate of 0.05. DA-TransUNet

Table 2: Results on Synapse multi-organ CT.

Method	DSC (%) ↑	HD95 ↓	Aorta	Gallbladder	Kidney (L)	Kidney (R)	Liver	Pancreas	Spleen	Stomach
CNN-based (reported)										
U-Net [1]	76.85	39.70	89.07	**69.72**	77.77	68.60	93.43	53.98	86.67	75.58
U-Net++ [2]	76.91	36.93	88.19	68.89	81.76	75.27	93.01	58.20	83.44	70.52
Attention U-Net [4]	77.77	36.02	**89.55**	68.88	77.98	71.11	93.57	58.04	87.30	75.75
MultiResUNet [12]	77.42	36.84	87.73	65.67	82.08	70.43	93.49	60.09	85.23	75.66
Transformer-based (reported)										
TransUNet [3]	77.48	31.69	87.23	63.13	81.87	77.02	94.08	55.86	85.08	75.62
UCTransNet [29]	78.23	26.75	84.25	64.65	82.35	77.65	94.36	58.18	84.74	79.66
TransNorm [30]	78.40	30.25	86.23	65.10	82.18	78.63	94.22	55.34	89.50	76.01
MIM [31]	78.59	26.59	87.92	64.99	81.47	77.29	93.06	59.46	87.75	76.81
Swin-Unet [14]	79.13	**21.55**	85.47	66.53	83.28	79.61	94.29	56.58	**90.66**	76.60
DA-TransUNet [6]	79.80	23.48	86.54	65.27	81.70	80.45	94.57	61.62	88.53	79.73
Retrained in this work (three seeds)										
DA-TransUNet [6]	80.26 ± 0.66	23.49	85.84	63.32	**83.29**	**80.64**	**94.89**	62.41	90.48	**81.16**
ApproxDA-TransUNet (Ours)	**80.57 ± 0.19**	24.52	88.13	66.23	82.04	79.88	94.67	**63.36**	89.19	81.08

Reported values as compiled by Sun et al. [6] (single runs). Retrained: three seeds (1234 / 0 / 1), final epoch; DSC as mean ± standard deviation, per-organ DSC as three-seed means. HD95 in voxels (TransUNet evaluation code, without voxel spacing). Bold: best; underlined: second best.

and ApproxDA-TransUNet are trained under identical settings. Every model is evaluated at the *final training epoch*, without checkpoint selection. Volume metrics are computed per case and averaged over classes; on FeTA, classes absent from a ground-truth volume are excluded from the average, as in nnU-Net.

4.2. Results on Each Benchmark

Tables 2–5 share one layout, following DA-TransUNet [6]. The upper rows list baselines, grouped into CNN-based and Transformer-based methods: values reported in the literature for Synapse, ACDC, CVC-ClinicDB, and ISIC 2018, and, because no comparable published 2D results exist for FeTA 2.4, baselines that we reproduced under our protocol. Synapse and ACDC use the standard splits of TransUNet (18/12 CT volumes; 70/10/20 patients), so their reported values are directly comparable; for CVC-ClinicDB and ISIC 2018 the splits differ (Table 4). The lower rows give DA-TransUNet and ApproxDA-TransUNet, both trained by us under identical settings with three seeds and evaluated at the final epoch. The comparison with the retrained DA-TransUNet is paired by seed; per-dataset p-values come from a paired t-test over the three seeds ($n = 3$, low power) and are given for completeness.

Synapse. ApproxDA-TransUNet reaches 80.57±0.19% Dice (Table 2), above all reported values, including those of DA-TransUNet (79.80%) and Swin-Unet (79.13%). Our retrained DA-TransUNet reaches 80.26%, close to its

Table 3: Results on ACDC cardiac MRI.

Method	**DSC (%) ↑**	RV	Myo	LV
CNN-based (reported)				
R50 U-Net† [1]	87.55	87.10	80.63	94.92
R50 Attention U-Net† [4]	86.75	87.58	79.20	93.47
Transformer-based (reported)				
R50 ViT† [13]	87.57	86.07	81.88	94.75
TransUNet [3]	89.71	88.86	84.53	95.73
Swin-Unet [14]	90.00	88.55	85.62	**95.83**
Retrained in this work (three seeds)				
DA-TransUNet [6]	90.94 ± 0.24	88.89	88.53	95.38
ApproxDA-TransUNet (Ours)	**91.08 ± 0.12**	**89.06**	**88.65**	95.54

Reported values from Cao et al. [14]; †ResNet-50 encoder variants. Retrained: as in Table 2. RV: right ventricle; Myo: myocardium; LV: left ventricle.

reported 79.80%, which supports the fairness of our training setup. Against it, ApproxDA-TransUNet is 0.32 points higher ($p = 0.43$) with a smaller seed standard deviation (0.19 vs. 0.66). The gain comes from the aorta (+2.29), gallbladder (+2.91), and pancreas (+0.95), whereas both kidneys (−1.26, −0.76) and the spleen (−1.29) are lower. HD95 is 24.52 voxels, against 23.49 for the retrained DA-TransUNet and 21.55 reported for Swin-Unet.

ACDC. ApproxDA-TransUNet reaches 91.08 ± 0.12% Dice (Table 3), above the values reported for Swin-Unet (90.00%) and TransUNet (89.71%). Against the retrained DA-TransUNet it is 0.15 points higher ($p = 0.55$), with all three structures slightly higher (+0.12 to +0.16).

CVC-ClinicDB and ISIC 2018. On CVC-ClinicDB (Table 4), ApproxDA-TransUNet reaches 89.88% Dice and 85.19% IoU against 89.47% and 84.55% for the retrained DA-TransUNet (+0.41 Dice, $p = 0.20$). On ISIC 2018 it is 0.20 points higher (88.51% vs. 88.30%, $p = 0.29$), with similar IoU (81.63% vs. 81.54%). Both retrained models are slightly below the ISIC 2018 values reported for TransUNet and DA-TransUNet, which were obtained on a different split.

FeTA 2.4. Table 5 places ApproxDA-TransUNet among six 2D models trained under the same protocol:

Table 4: Results on CVC-ClinicDB and ISIC 2018.

Method	**CVC-ClinicDB**		**ISIC 2018**	
	DSC (%) ↑	IoU (%) ↑	DSC (%) ↑	IoU (%) ↑
CNN-based (reported)				
U-Net [1]	86.93	78.21	87.22	81.14
U-Net++ [2]	87.14	78.47	87.30	81.33
Attention U-Net [4]	87.41	79.35	87.60	81.51
Transformer-based (reported)				
TransUNet [3]	89.01	81.63	88.78	82.63
DA-TransUNet [6]	89.47	82.51	**88.88**	**82.78**
Retrained in this work (three seeds)				
DA-TransUNet [6]	89.47 ± 0.23	84.55	88.30 ± 0.20	81.54
ApproxDA-TransUNet (Ours)	**89.88 ± 0.48**	**85.19**	88.51 ± 0.28	81.63

Reported values from Table 2 of Sun et al. [6], obtained on random 75/25 splits that were not released; our 75/25 splits differ, so the comparison with these rows is indicative only. Retrained: as in Table 2.

- **CNN and Swin baselines**: ApproxDA-TransUNet (81.05%) is above U-Net (+1.85 points, $p = 0.018$), U-Net++ (+2.04, $p = 0.021$), Attention U-Net (+1.94, $p = 0.073$), and Swin-Unet (+2.46, $p = 0.059$).
- **Transformer hybrids**: it is 0.07 points below TransUNet, a small but consistent gap (lower for all three seeds, $p = 0.036$), with a tenth of its parameters (10.75M vs. 105.28M) and under a quarter of its FLOPs (6.9 vs. 29.3 GFLOPs), and 0.80 points above DA-TransUNet ($p = 0.17$).
- **Tissues**: relative to DA-TransUNet, the gains are in grey matter (+2.58), white matter (+2.66), and external CSF (+1.67); the lateral ventricles, cerebellum, and deep grey matter are on par (+0.50, −0.12, −0.37), and the brainstem is lower (−1.34).
- **HD95**: ApproxDA-TransUNet has the highest HD95 of the seven models (2.85 mm, against 1.90 mm for TransUNet and 2.17 mm for DA-TransUNet).

Case Audit of FeTA HD95 (Sub-052 Outlier). The higher HD95 of ApproxDA-TransUNet on FeTA comes from a single subject, **sub-052**, which has severe ventriculomegaly (lateral-ventricle to white-matter volume ratio 2.29, against a training median of 0.16 and a training maximum of 1.81) and is therefore

Table 5: Results on FeTA 2.4 fetal brain MRI.

Method	**DSC (%) ↑**	**HD95 ↓**	eCSF	GM	WM	LV	CBM	SGM	BS
CNN-based (reproduced, three seeds)									
U-Net [1]	79.21 ± 0.34	2.02	73.75	67.74	84.40	84.09	**82.49**	83.08	78.90
U-Net++ [2]	79.01 ± 0.16	2.13	73.26	67.41	84.10	83.82	82.28	83.29	78.94
Attention U-Net [4]	79.11 ± 0.34	2.08	73.50	66.84	84.45	84.38	82.25	83.49	78.86
Transformer-based (reproduced, three seeds)									
TransUNet [3]	**81.12 ± 0.63**	**1.90**	74.53	67.99	87.73	**88.95**	82.23	**85.02**	**81.41**
Swin-Unet [14]	78.60 ± 0.46	2.60	74.13	68.63	83.95	83.60	81.12	84.74	74.01
Retrained in this work (three seeds)									
DA-TransUNet [6]	80.26 ± 0.66	2.17	73.86	68.70	85.25	87.43	81.89	84.07	80.60
ApproxDA-TransUNet (Ours)	81.05 ± 0.63	2.85	**75.53**	**71.29**	**87.91**	87.93	81.76	83.69	79.25

All models trained under the same protocol (Swin-Unet with its official learning rate of 0.05): three seeds (1234 / 0 / 1), final epoch; DSC as mean ± standard deviation, per-tissue DSC as three-seed means. Test set of 8 subjects; HD95 in millimeters. eCSF: external cerebrospinal fluid; GM: grey matter; WM: white matter; LV: lateral ventricles; CBM: cerebellum; SGM: deep grey matter; BS: brainstem.

outside the training distribution. On this subject, ApproxDA-TransUNet predicts a few small cerebellum regions (the largest has 34, 170, and 266 voxels for the three seeds) about 50 mm away from the true cerebellum, which raises the cerebellum HD95 to 28–50 mm and the subject's mean HD95 to 8.88, 8.89, and 13.65 mm (DA-TransUNet: 2.85, 4.43, 4.02 mm). Excluding sub-052, the mean HD95 over the other seven subjects is 1.76 mm for ApproxDA-TransUNet and 1.94 mm for DA-TransUNet.

4.3. Paired Comparison Across Datasets

Across the five benchmarks, the three-seed mean Dice of ApproxDA-TransUNet is higher than that of the retrained DA-TransUNet on all five datasets. The per-seed differences (seeds 1234 / 0 / 1) are +0.87 / −0.26 / +0.34 points on Synapse, +0.55 / −0.004 / −0.11 on ACDC, +0.60 / −0.03 / +0.65 on CVC-ClinicDB, +0.35 / +0.33 / −0.08 on ISIC 2018, and +0.73 / +1.47 / +0.19 on FeTA 2.4: 10 of the 15 seed pairs are positive, and the average gain is +0.37 points. None of the per-dataset differences is significant with three seeds. Because the three seeds of a dataset are not independent experiments, we treat the five datasets as the independent experimental units: the mean dataset-level gain is +0.37 points (one-sample t-test across the five dataset means, $p = 0.031$), and a hierarchical bootstrap that resamples datasets and then seeds yields a 95% interval of $[0.13, 0.68]$ points. With only five datasets these estimates are themselves uncertain, and we read them as evidence that accuracy is retained rather than as a claim of

superiority. HD95, reported for Synapse and FeTA 2.4, does not follow the Dice trend: it is higher than DA-TransUNet's on both (significantly only on FeTA, $p = 0.04$, where the difference comes from sub-052).

4.4. Qualitative Visual Evaluation

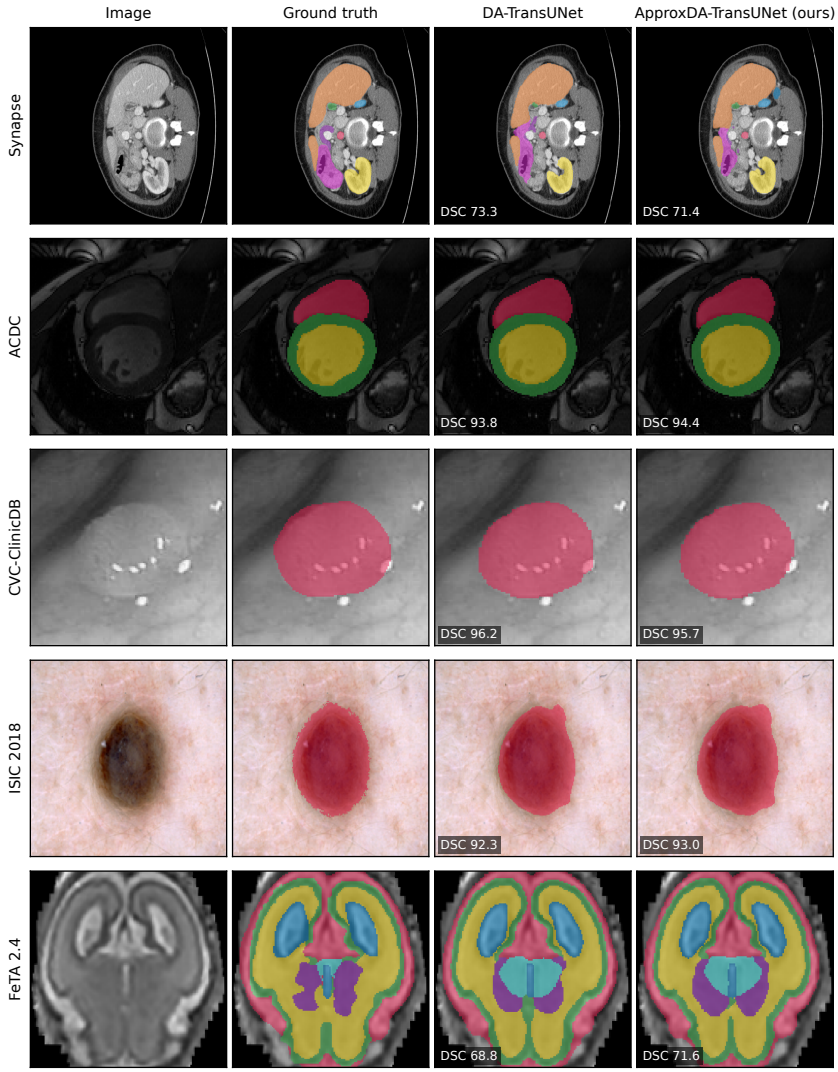


Figure 5: Qualitative results on the five datasets.

Figure 5 shows one representative example per dataset (seed 1234, final epoch). The example is chosen without reference to DA-TransUNet: the test

Table 6: Single-axis ablations on Synapse.

Ablation Variant	Mean DSC (%)	HD95	Paired Diff. (p)
ApproxDA-TransUNet baseline	**80.57 ± 0.19**	**24.52**	–
Block presence and exact dual attention			
w/o ApproxDA blocks	77.82 ± 0.33	29.12	**−2.76** (0.008)
Exact global dual attention (M=112, r=0, G=1)	80.56 ± 0.62	23.64	−0.01 (0.97)
Window size M			
$M = 112$ (global window at every stage)	80.07 ± 0.98	28.99	−0.50 (0.54)
Projection rank r			
$r = 0$ (exact attention inside each window)	80.76 ± 0.15	21.12	+0.19 (0.44)
$r = 64$	79.64 ± 0.40	26.85	−0.93 (0.11)
Channel groups G			
$G = 1$ (un-grouped CAM)	81.12 ± 0.77	23.48	+0.55 (0.25)
$G = 4$	79.91 ± 0.57	26.69	−0.67 (0.10)
Encoder			
R50 third stage instead of LightViT-T	78.81 ± 0.27	27.71	**−1.76** (0.007)
ViT-B/16 instead of LightViT-T	78.52 ± 1.64	26.01	−2.05 (0.14)
LightViT-S instead of LightViT-T	80.31 ± 0.63	23.62	−0.26 (0.43)
Bottleneck placement and decoder			
Bottleneck block before LightViT-T	79.47 ± 0.36	26.10	**−1.10** (0.008)
Full decoder ($256 \to 128 \to 64 \to 16$)	79.60 ± 1.64	25.63	−0.97 (0.40)

Seeds 1234 / 0 / 1, 224×224, 9 classes. Baseline: ApproxDA-TransUNet ($M = 28$, $r = 32$, $G = 8$, $g = \frac{1}{2}$, LightViT-T, slim decoder). HD95 in voxels. Paired differences and p-values (paired t-test over three seeds, $n = 3$) are relative to the baseline.

case with the median Dice of ApproxDA-TransUNet and, for volumes, the slice with the most annotated classes; panels are cropped to the annotated region, and the Dice of the displayed slice or image is given on each prediction. On these median cases the two models produce similar masks: ApproxDA-TransUNet is slightly better on ACDC, ISIC 2018, and FeTA 2.4 and slightly worse on Synapse and CVC-ClinicDB. The visible differences are small and lie mostly along structure boundaries.

5. Ablation Studies

Single-axis ablations are run on Synapse with three seeds (1234, 0, 1). The reference is ApproxDA-TransUNet (80.57 ± 0.19% Dice, HD95 24.52 voxels); each variant changes exactly one factor.

5.1. Dual-Attention Blocks and Their Approximation

- **The blocks matter in this configuration**: removing all ApproxDA blocks lowers Dice on Synapse by 2.76 points for every seed ($p = 0.008$) and raises HD95 from 24.52 to 29.12; the largest drops are in the gallbladder and pancreas.

- **The approximation costs no accuracy**: replacing the approximated attention by the exact global dual attention of DA-TransUNet ($M = 112$, $r = 0$, $G = 1$) leaves Dice unchanged (-0.01, $p = 0.97$), and exact attention inside each window ($r = 0$) is also on par ($+0.19$). The exact global variant costs 14% more FLOPs and about twice the training time per run (94 vs. 45 minutes).

5.2. Approximation Hyperparameters

- **Window size and rank**: a global window ($M = 112$, -0.50, $p = 0.54$) and a higher rank ($r = 64$, -0.93, $p = 0.11$) are not significantly different from the defaults; with three seeds on Synapse, differences below about one point cannot be resolved.

- **Channel groups**: $G = 8$ is a fixed default carried over from the ViT-B/16 version of the block, where the bottleneck has $C' = 48$ channels and each group six; it was not tuned for the lightweight model. Here it leaves one channel per group in the bottleneck ($C' = 8$) and the 1/2 skip ($C' = 4$, four groups), where CAM degenerates to the scaling $(1 + \beta)\mathbf{X}'$ (Section 3). Un-grouped CAM ($G = 1$) is 0.55 points higher ($p = 0.25$) at no extra cost. We nevertheless keep $G = 8$ in the final model as a fixed, non-tuned setting, for consistency with the workshop design and with all other experiments reported here; the results do not show that $G = 8$ is optimal, and $G = 1$ or a stage-adaptive grouping (Section 6) is a reasonable alternative.

- **Channel groups** $G = 4$: -0.67 ($p = 0.10$), not significant.

- **Encoder**: replacing LightViT-T with the third ResNet-50 stage (the TransUNet stem; 15.42M parameters in total) lowers Dice by 1.76 points for every seed ($p = 0.007$) while adding 4.7M parameters. ViT-B/16 is also lower (-2.05, $p = 0.14$), and the larger LightViT-S is not better (-0.26, $p = 0.43$), so the lightest transformer tested is sufficient.

- **Block placement**: applying the bottleneck block before LightViT-T, on the $28 \times 28 \times 512$ stage-2 output as in DA-TransUNet's ordering, lowers Dice by 1.10 points for every seed ($p = 0.008$) and adds 0.31M parameters. The full decoder of DA-TransUNet is not better than the slim decoder (-0.97, $p = 0.40$) at 1.3 times the parameters and 1.5 times the FLOPs.

6. Discussion

This section discusses possible left–right errors from mirror augmentation, reports deployment costs, and outlines limitations.

6.1. Anatomical Symmetries and Data Augmentation

In an error analysis on the Synapse test set (with the earlier ViT-B/16 model), most misclassified voxels were boundary or extent errors between an organ and the background. The confusions between organs were mostly whole-organ misidentifications, chiefly stomach versus spleen and left versus right kidney, and were concentrated in a few test cases.

The left–right confusions are a plausible side effect of the standard TransUNet augmentation: `random_rot_flip` follows a random 90° rotation with an axis flip but does not swap the left and right labels, so about half of the augmented slices show mirrored anatomy with unchanged laterality labels. We did not quantify this effect on the reported benchmarks; laterality-aware augmentation, which swaps the labels of bilateral structures when an image is mirrored, is a simple safeguard for such structures.

6.2. Computational Efficiency and Deployment Profile

The lower part of Table 1 reports the cost measured on the GPU. ApproxDA-TransUNet needs 58% less GPU memory for training and inference, and its FP32 checkpoint is ten times smaller (43.5 vs. 432 MB). Because volumes are segmented slice by slice in batches, throughput is the relevant speed measure, and it is 2.55 times higher. At batch size 1, however, the lightweight model is slower (9.6 vs. 6.5 ms): its many small LightViT and attention kernels leave the GPU underutilized, so the FLOP reduction does not translate into lower single-slice latency.

6.3. Limitations and Future Work

The channel grouping is a fixed default ($G = 8$) rather than an optimized design: with the small compressed widths of the lightweight model it removes cross-channel interaction in two of the four blocks. A stage-adaptive grouping that keeps at least two channels per group (for example $G = C'/4$) would be the more principled choice. The ablations were run on Synapse only, where three seeds resolve differences of about one Dice point at best, and the statistical analysis across datasets rests on five benchmarks. ApproxDA-TransUNet processes 3D scans on a 2D planar basis. It therefore cannot use inter-slice context, which matters, for example, at the cranial and caudal ends of organs. Extending ApproxDA blocks to lightweight 2.5D or anisotropic 3D formulations presents a promising avenue for volumetric research.

7. Conclusion

In this paper, we presented **ApproxDA-TransUNet**, a lightweight redesign of DA-TransUNet that combines a truncated ResNet-50 stem, LightViT-T, approximated dual-attention blocks (windows of $M = 28$, rank $r = 32$, and a fixed, non-tuned channel grouping) on the bottleneck and all skips, and a slim decoder. It uses 90.0% fewer parameters (10.75M vs. 107.95M) and 77.1% fewer FLOPs (6.92 vs. 30.21 GFLOPs). Over three seeds on five benchmarks (Synapse, ACDC, CVC-ClinicDB, ISIC 2018, and FeTA 2.4), its Dice is comparable to that of DA-TransUNet (higher three-seed mean on all five datasets, average gain +0.37 points), while HD95 is higher on both datasets for which it is reported; on FeTA 2.4 it is within 0.1 points of TransUNet. In deployment it needs 58% less GPU memory and processes 2.55 times more slices per second at batch size 32, although single-slice latency at batch size 1 is higher (9.6 vs. 6.5 ms).

In single-axis ablations on Synapse, removing the dual-attention blocks reduces Dice by 2.76 points, and windowed low-rank position attention and grouped channel attention match the exact global dual attention of DA-TransUNet at lower cost (the exact variant needs 14% more FLOPs and twice the training time), showing that, in the evaluated configuration, dual attention can be retained without its full quadratic overhead.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work

reported in this paper.

CRediT Authorship Contribution Statement

Author One: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing - Original Draft, Visualization. **Author Two**: Software, Validation, Data curation, Writing - Review & Editing. **Author Three**: Supervision, Project administration, Resources, Funding acquisition, Writing - Review & Editing.

Data and Code Availability

All datasets used in this study (Synapse BTCV, ACDC, CVC-ClinicDB, ISIC 2018, and FeTA 2.4) are publicly accessible benchmark repositories. The complete source code and pretrained model checkpoints for ApproxDA-TransUNet will be made publicly available upon publication.

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